

Program: Diploma in Civil and Environmental Engineering	
Course Code: 4358	Course Title: Quantity Surveying Lab
Semester: 4	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0 T:0 P:3)	Periodspersemester: 45

Course Objectives:

- To impart the knowledge about the procedure for estimating and costing of Civil Engineering works
- To develop in the student the skill in the use of spreadsheet for the preparation of abstract of estimate for construction projects.
- To impart the ability to prepare detailed and abstract estimate for various construction projects.
- To prepare valuation certificates and statements of a real and a landed property.

Course Prerequisites:

Topic	Course code	Course name	Semester
Elementary Mathematics		Engineering mathematics	1&2
Civil Engineering Drafting Lab		Building Drawing and Estimation	3
Building construction		Building materials and construction	3

Course Outcomes:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Identify the basic principles for estimation and costing of Civil Engineering works	12	Applying
CO2	Prepare detailed estimate for infrastructure works using spreadsheet	13	Applying
CO3	Prepare abstract of estimate for construction projects using spreadsheet.	9	Applying

CO4	Prepare valuation certificates and statements of a real and a landed property and fix rent of properties existing government building	7	Applying
	Lab Tests	4	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2				3			
CO3				3			
CO4	3				3		

3-Stronglymapped, 2-Moderatelymapped,1-Weaklymapped

Course Outline

On completion of the course student will be able to:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Identify the basic principles for estimation and costing of Civil Engineering works.		
M1.01	Understand the purpose of estimation and the requirements of an estimator. Explain the rules of measurement as per IS:1200.	3	Understanding
M1.02	Prepare a report on market rates for given material, labour wages, hire charges of tools & Equipment.	3	Applying
M1.03	Prepare the list of items to be executed with units for detailed estimate of a given structure from the given drawing.	3	Applying
M1.04	Compute volume of earth work by different methods.	3	Applying

CO2	Prepare detailed estimate for infrastructure works using spreadsheet		
M2.01	Understand different methods of preparing Detailed Estimate.	2	Understanding
M2.02	Prepare detailed estimate of small Septic tank and soak pit from the given set of drawings.	3	Applying
M2.03	Prepare detailed estimate of road of one kilometer length from the given drawing (CC Road and WMM).	3	Applying
M2.04	Prepare detailed estimate and abstract of estimate of a flat roof and tiled roof building.	3	Applying
	Lab Test-I	2	
CO3	Prepare abstract of estimate for construction projects using spreadsheet		
M3.01	Understand the purpose and importance of rate analysis. Familiarize with Standard data book and Schedule of rates.	2	Understanding
M3.02	Prepare data and analysis of rates for various items of work.	3	Applying
M3.03	Prepare detailed estimate and abstract of estimate of septic tank, open well, water tank, compound wall, road work.	4	Applying
CO4	Prepare valuation certificates and statements of a real and a landed property and fix rent of properties- existing government buildings.		
M4.01	Open ended experiment- Detailed estimate and abstract of estimate, valuation certificate, Fix the rent.	7	Applying
	Lab Test II	2	

Text/Reference:

T/R	BookTitle/Author
T1	Datta, B.N., Estimating and Costing in Civil engineering, UBS Publishers Distributors Pvt. Ltd. New Delhi.
R1	Rangwala,S.C., Estimating and Costing, Charotar Publishing House PVT. LTD., Anand
R2	Birdie,G.S., Estimating and Costing, Dhanpat Rai Publishing Company(P) Ltd. NewDelhi
R3	V N Vazirani& S P Chandola, Civil engineering Estimating and Costing, Khanna Publishers.
R4	Chakraborti,M., Estimating and costing, specification and valuation in civil engineering, MonojitChakraborti, Kolkata
R5	IS 1200-1968; Methods of measurement of Building & Civil Engineering works.
R6	CPWD Schedule of Rates for Delhi-2018
R7	CPWD Analysis of Rates for Delhi-2018, Vol I and II
R8	Ministry of Road Transport and Highways (MORT&H) Specifications and Analysis of Schedule of Rates.
R9	IRC SP: 20 Rural Roads Manual

Online Resources:

Sl.No	Website Link
1	https://nptel.ac.in/
2	http://egyankosh.ac.in/
3	https://swayam.gov.in/